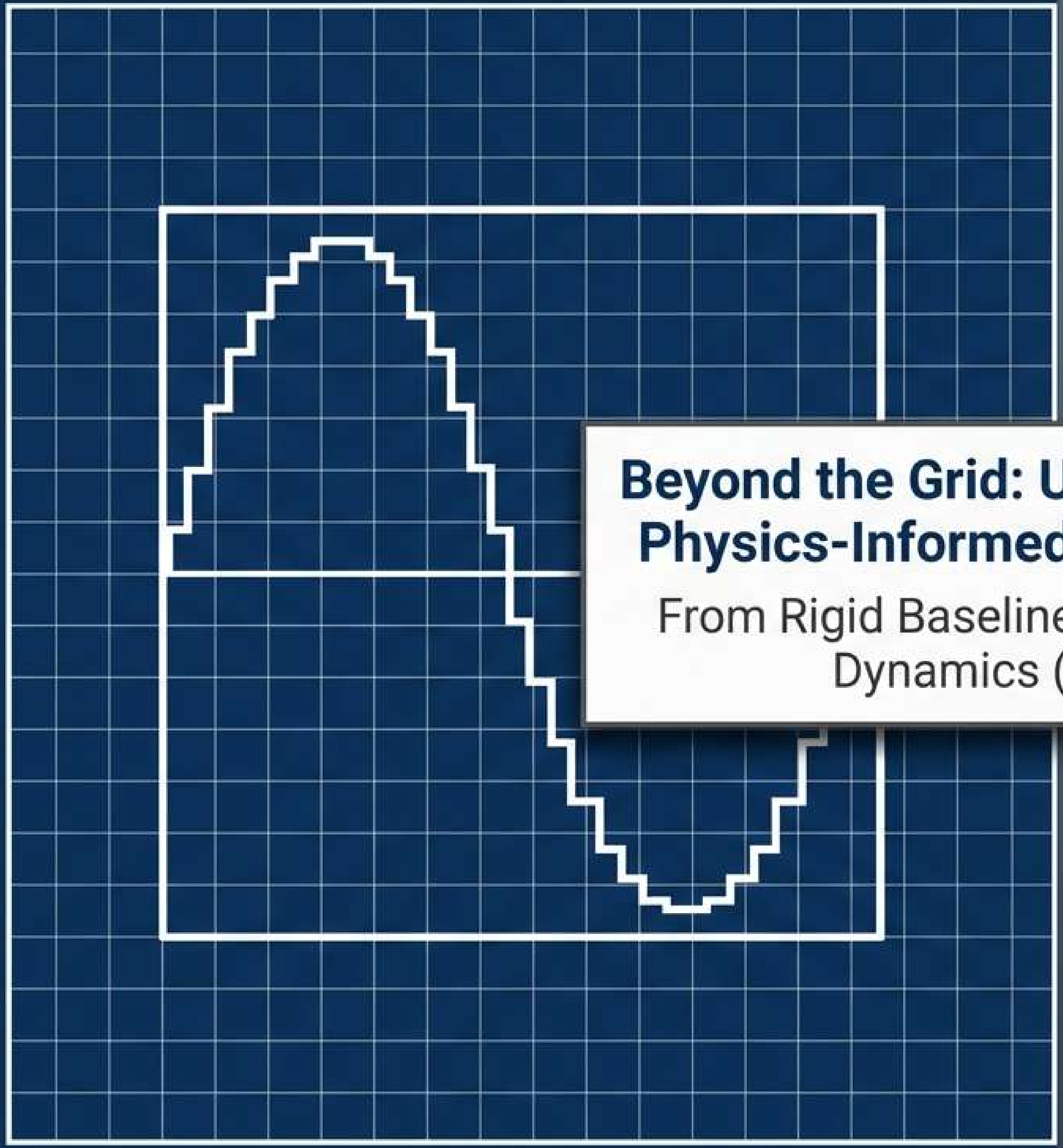
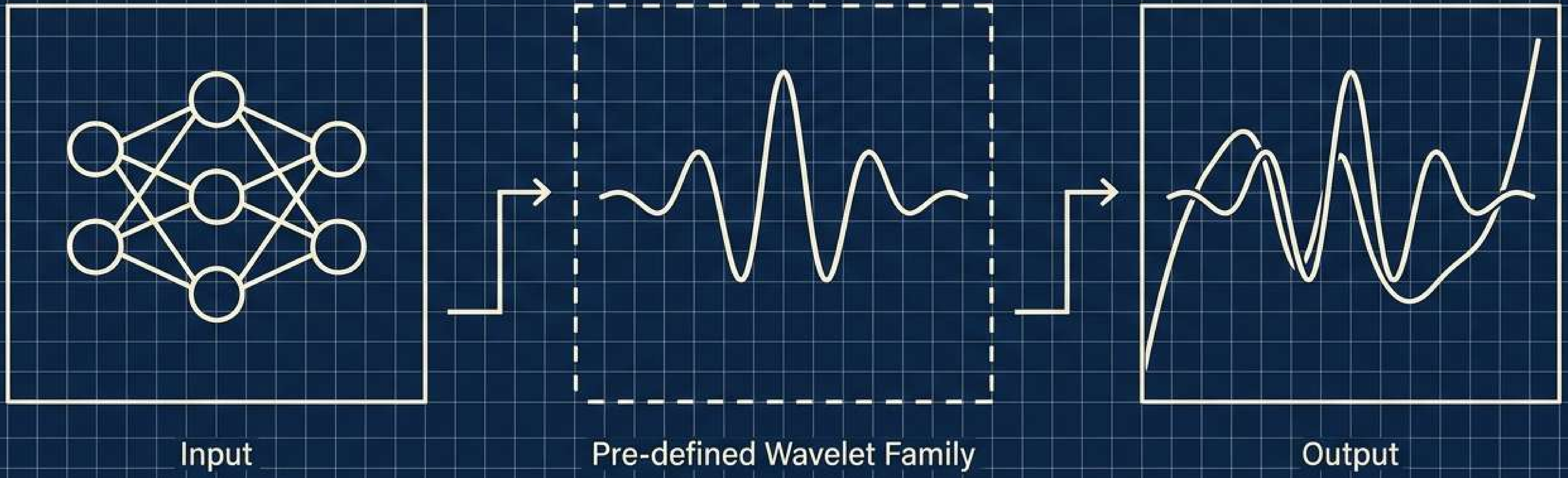


Beyond the Grid: Unlocking Adaptive Physics-Informed Neural Networks

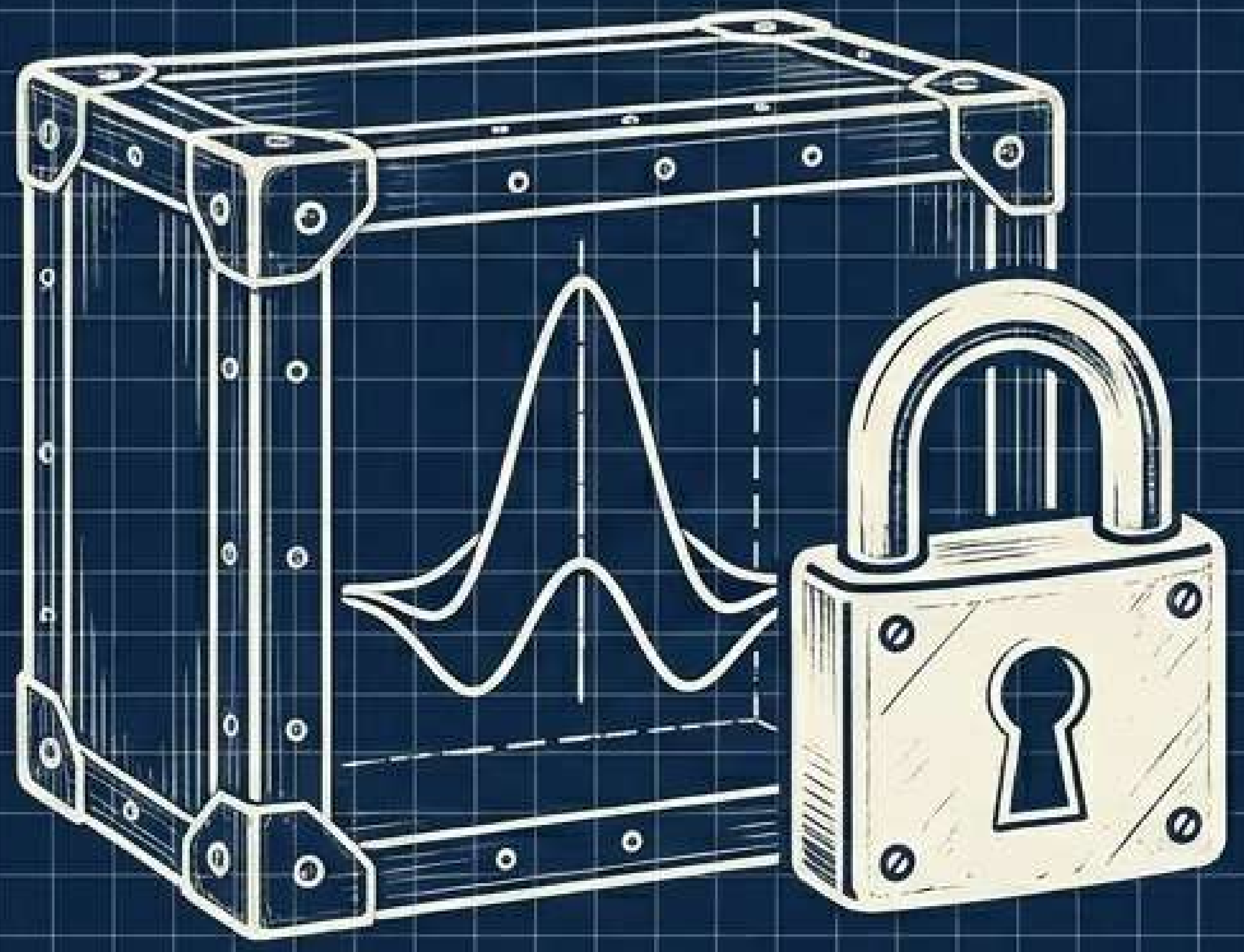
From Rigid Baselines to Meta-Wavelet Dynamics (MW-PINN)

Why Build with Wavelets?

Instead of forcing a neural network to solve differential equations from scratch, we supply it with pre-defined mathematical building blocks (wavelets). By combining these known shapes, the network approximates complex equations with analytical precision.

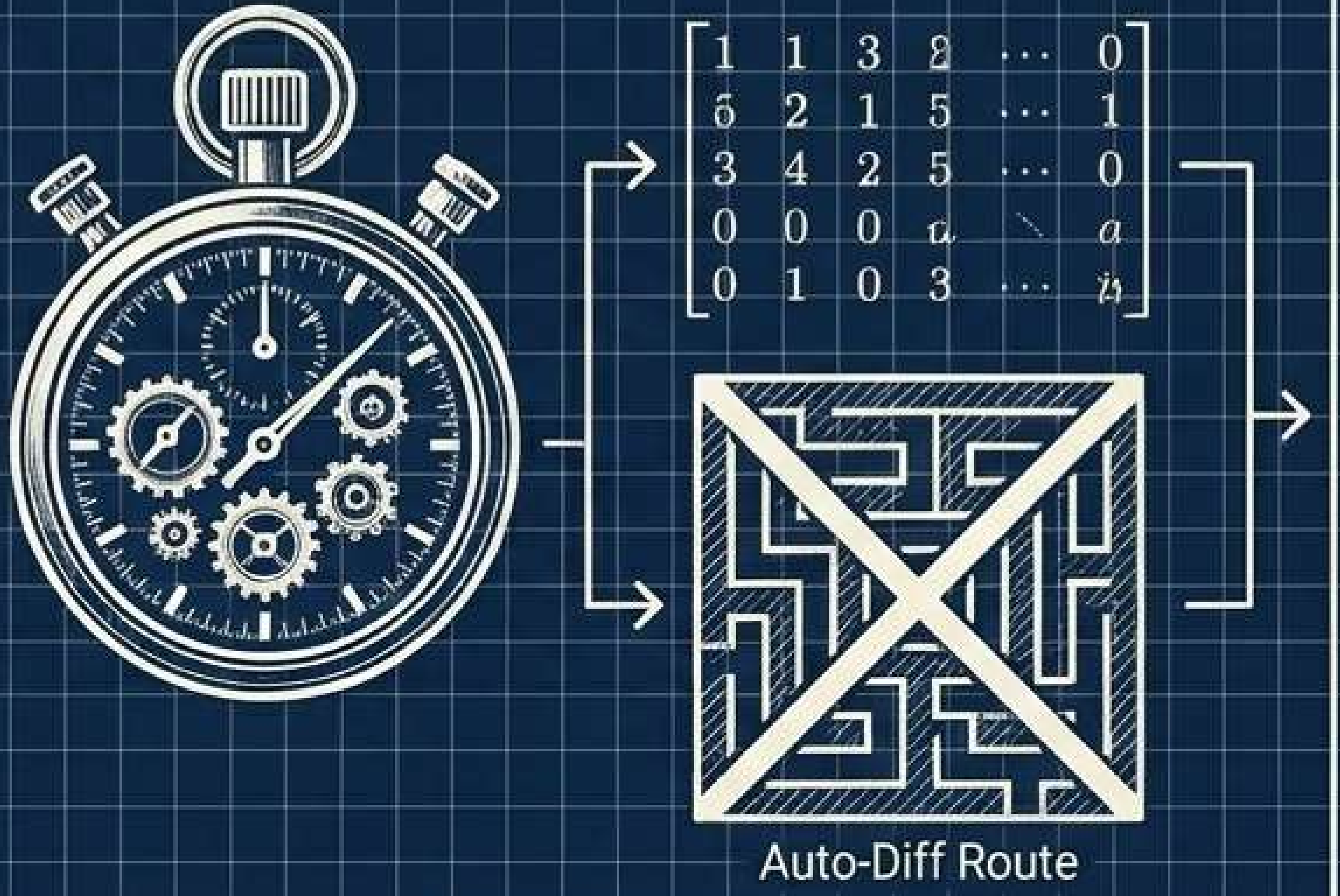


Wavelet PINN (W-PINN): The Rigid Foundation



Limitation

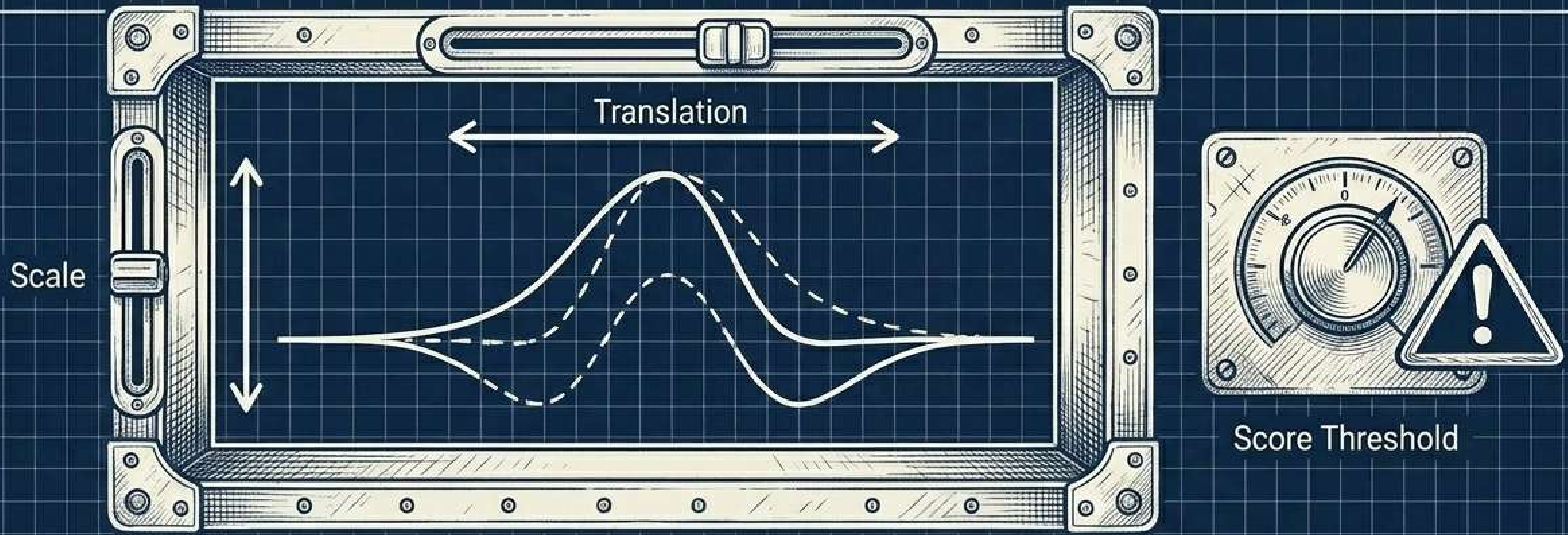
The wavelet shape is fixed forever. It relies on a pre-defined family (like the Gaussian or Mexican Hat wavelet) and cannot adapt.



Advantage

Precomputing derivative matrices analytically bypasses slow automatic differentiation. Training is blazing fast.

Adaptive Wavelet PINN (AW-PINN): The Incremental Upgrade



What it Does

Learns the scale and translation of each wavelet so they can slide around and resize to fit the problem.

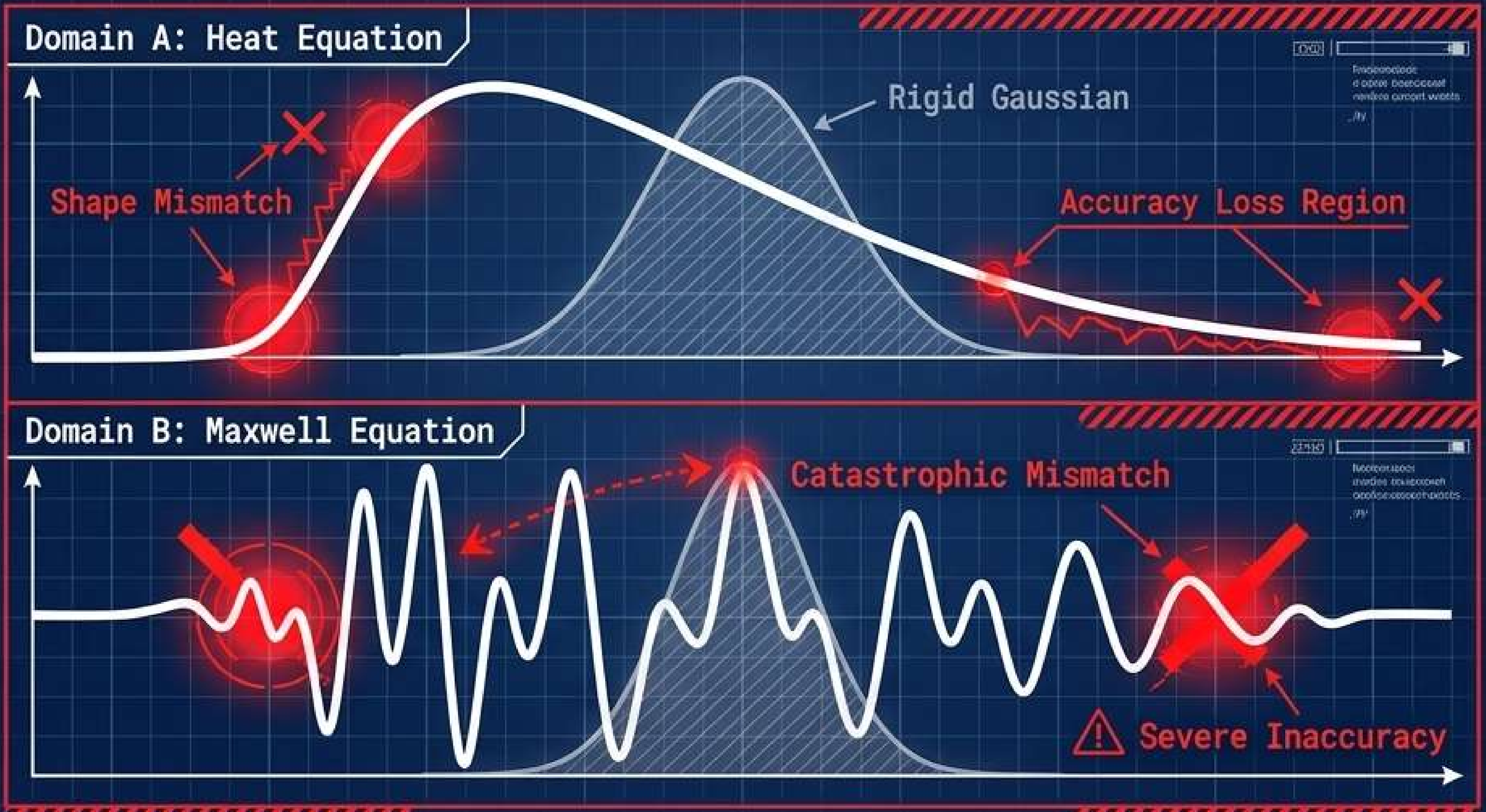
The Limitations

The underlying shape remains stubbornly fixed. Additionally, it forces users to manually guess a score threshold to determine which wavelets to keep.

The Core Problem: The Artificial Shape Constraint

The Constraint

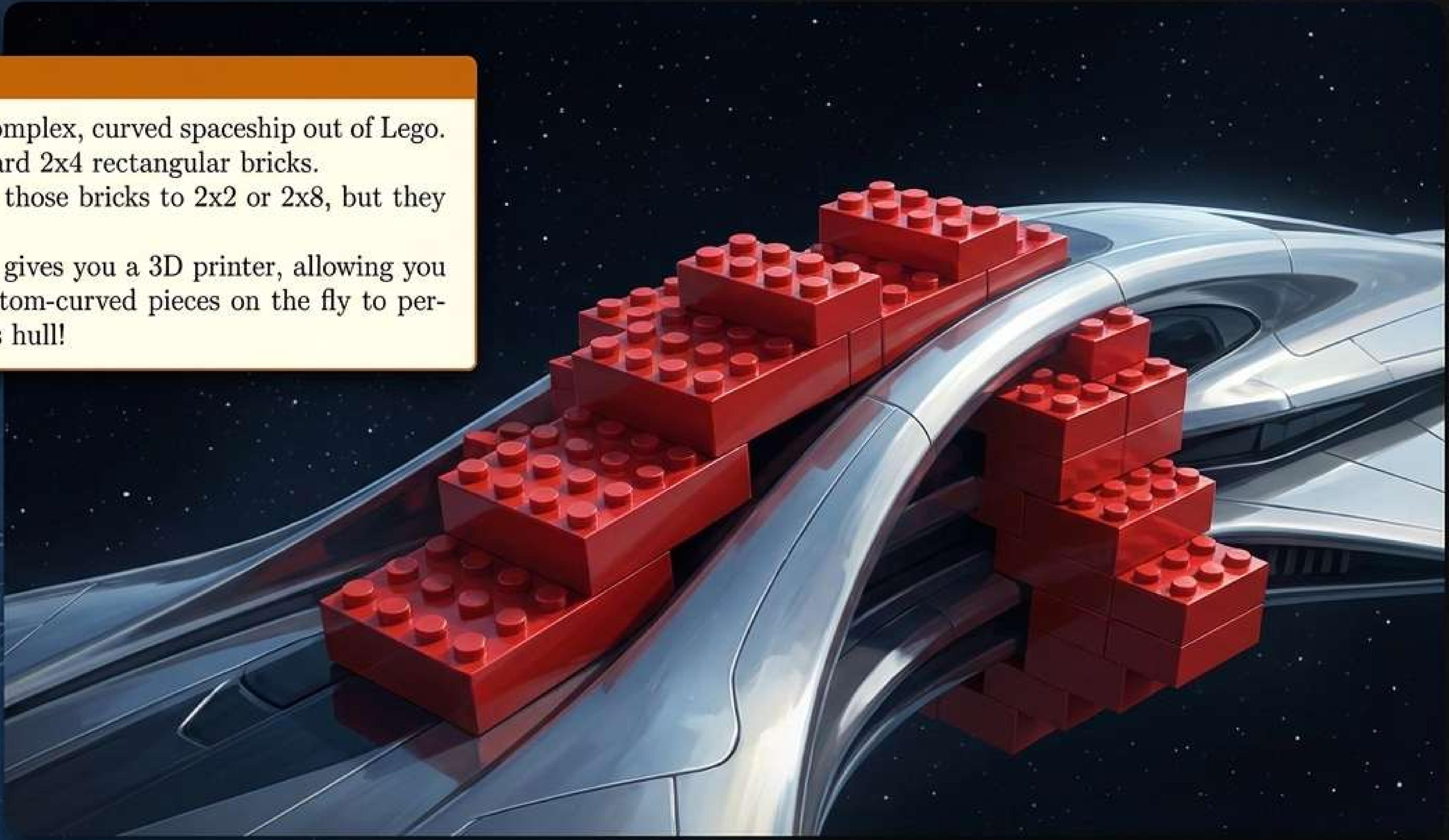
The optimal wavelet shape for a heat equation is NOT the same as the optimal shape for a Maxwell equation. Fixing the shape is an artificial constraint that hurts accuracy.



The Blueprint Constraint in the Physical World

The Lego Analogy

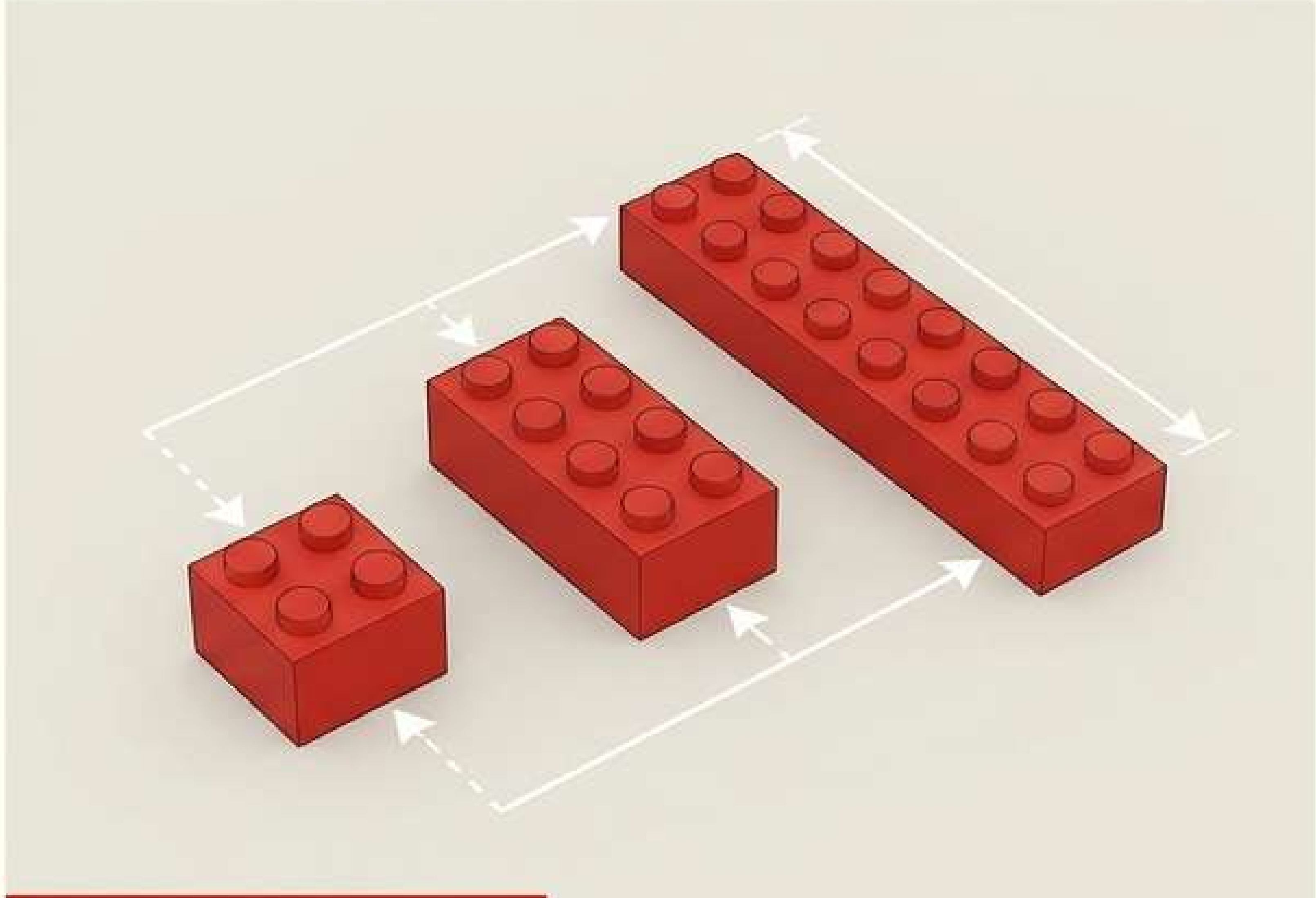
Imagine trying to build a complex, curved spaceship out of Lego.
W-PINN gives you standard 2x4 rectangular bricks.
AW-PINN lets you resize those bricks to 2x2 or 2x8, but they remain rectangles.
MW-PINN (our method) gives you a 3D printer, allowing you to invent entirely new, custom-curved pieces on the fly to per-
perfectly fit the spaceship's hull!



Constructing with Rigid Baselines

W-PINN

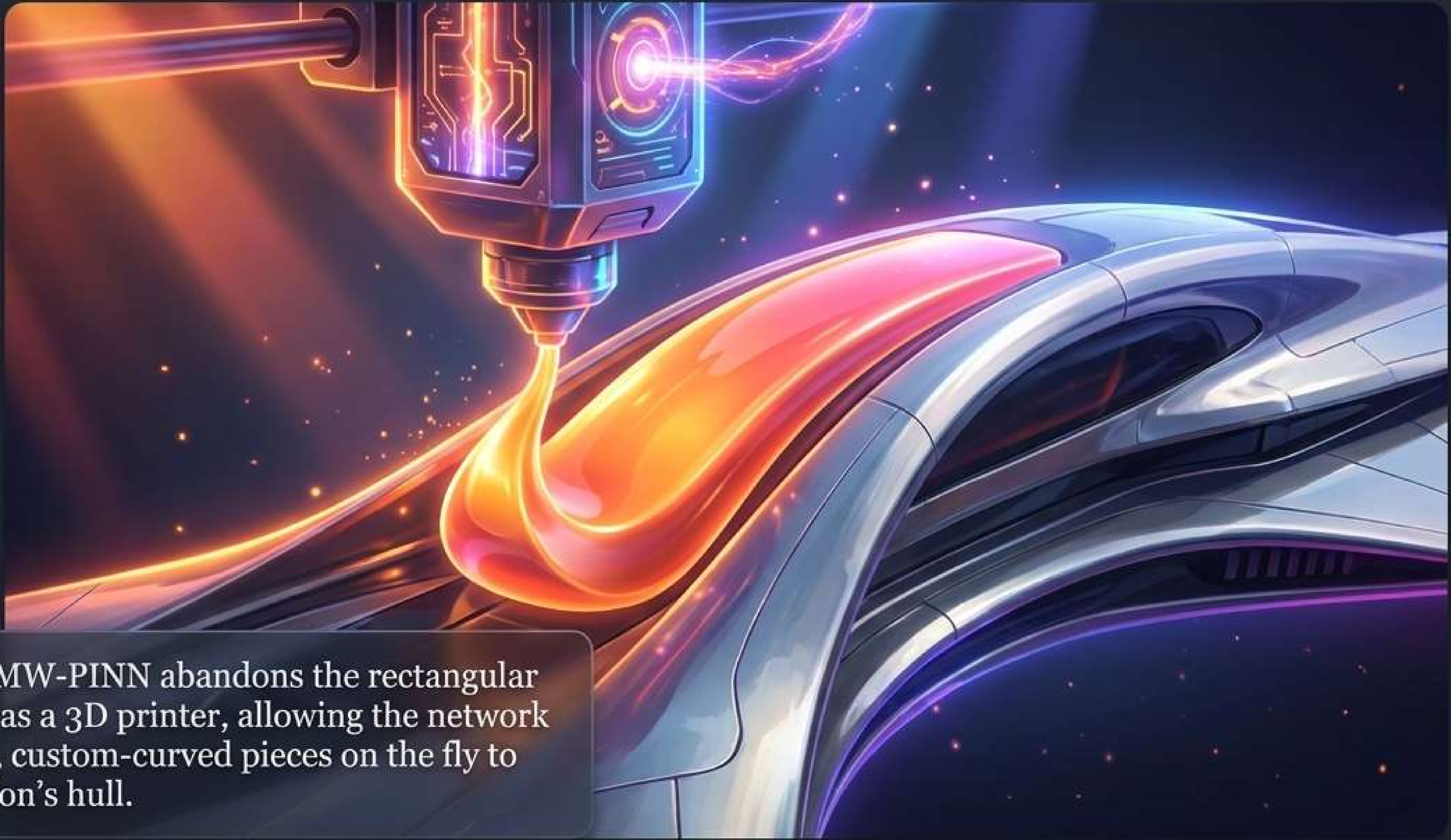
Gives you standard 2x4 rectangular bricks to build your curve.



AW-PINN

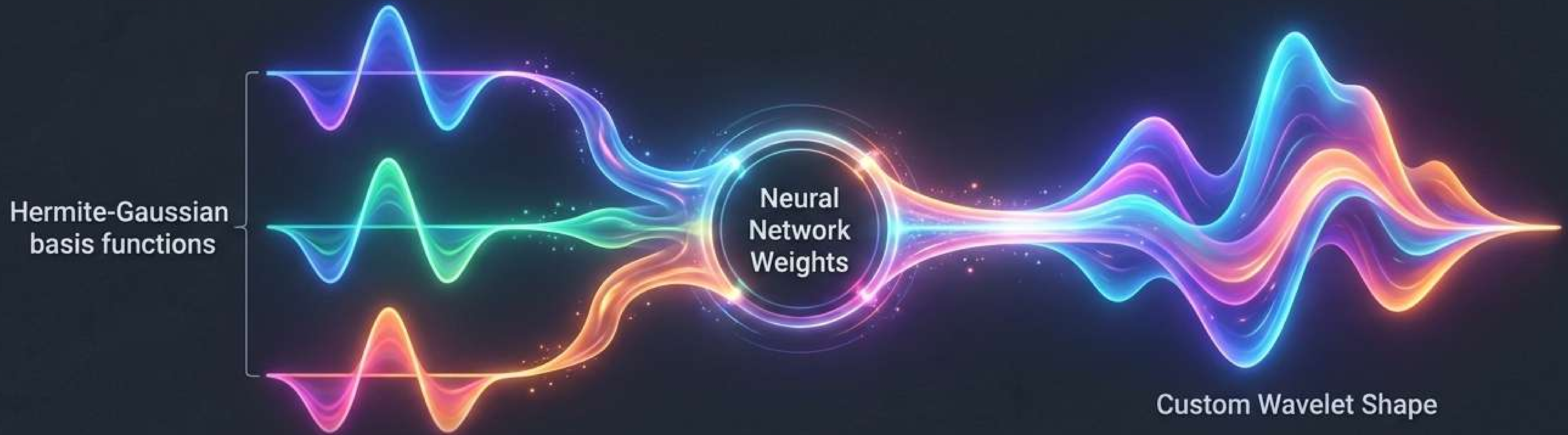
Lets you resize those bricks to 2x2 or 2x8, but they remain fundamentally rigid rectangles.

Meta-Wavelet PINN (MW-PINN): The 3D Printer



The Breakthrough: MW-PINN abandons the rectangular bricks entirely. It acts as a 3D printer, allowing the network to invent entirely new, custom-curved pieces on the fly to perfectly fit the equation's hull.

Engineering the Shape Space



Mechanism

Instead of a single fixed bump, the wavelet is defined as a learnable linear combination of Hermite-Gaussian basis functions.

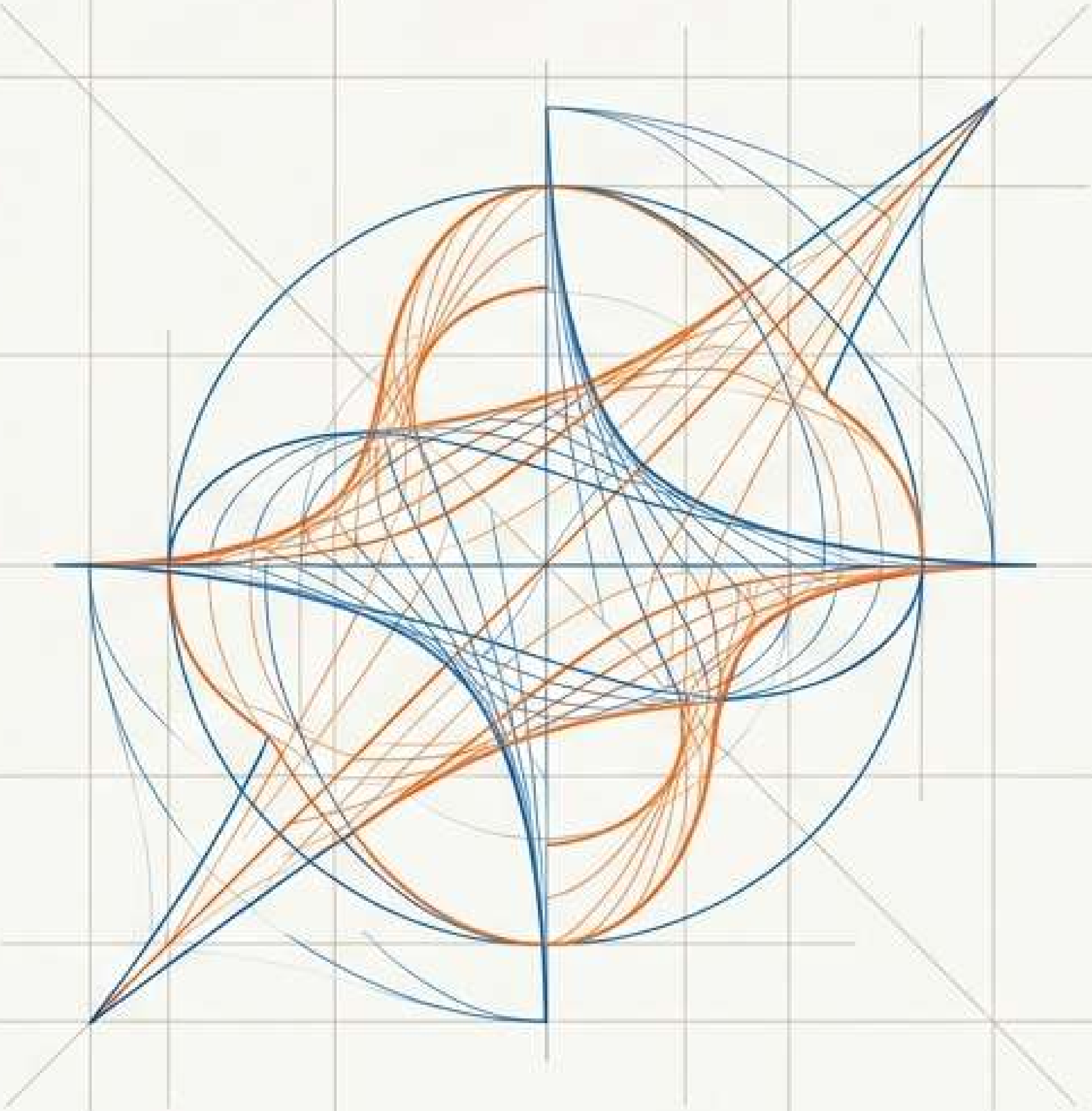
Result

The network freely mixes these parameters during training to discover any weighted mathematical combination, optimizing shape alongside scale and translation.

Minimum Compromise, Maximum Flexibility

	W-PINN	AW-PINN	MW-PINN
Learns Scale/Translate	No	Yes	Yes ✓
Learns Wavelet Shape	No	No	Yes ✓
Analytical Derivatives (Speed)	Yes	Yes	Yes ✓
Selection Mechanism	None	Manual Threshold	Auto (L1 Penalty) ✓

Takeaway: MW-PINN maintains the blazing speed of analytical derivatives while completely eliminating artificial shape constraints and manual threshold tuning.

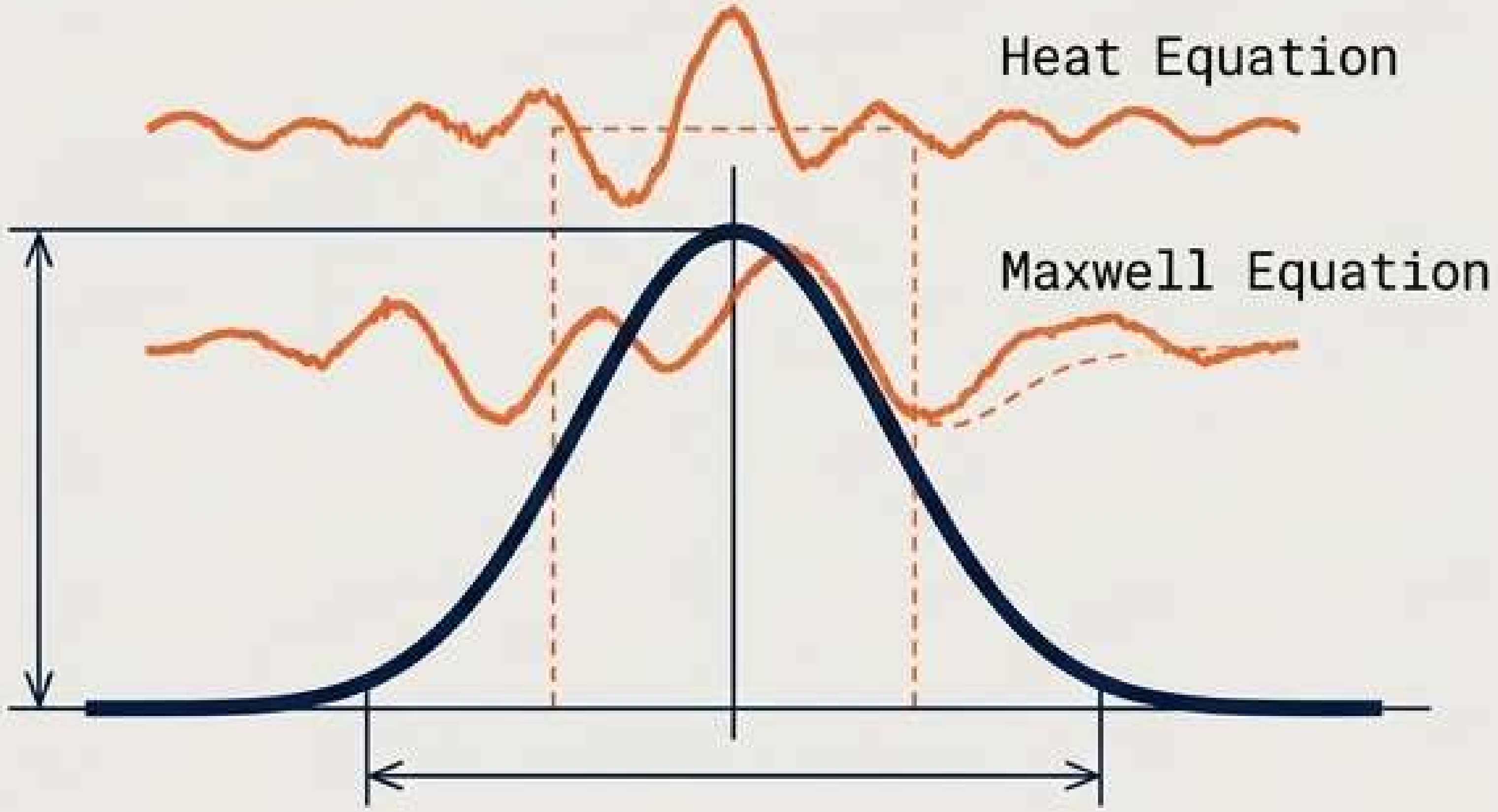


Meta-Wavelet PINN (MW-PINN)

A Dynamic Blueprint for Self-Optimizing Shape Spaces in Physics-Informed Neural Networks

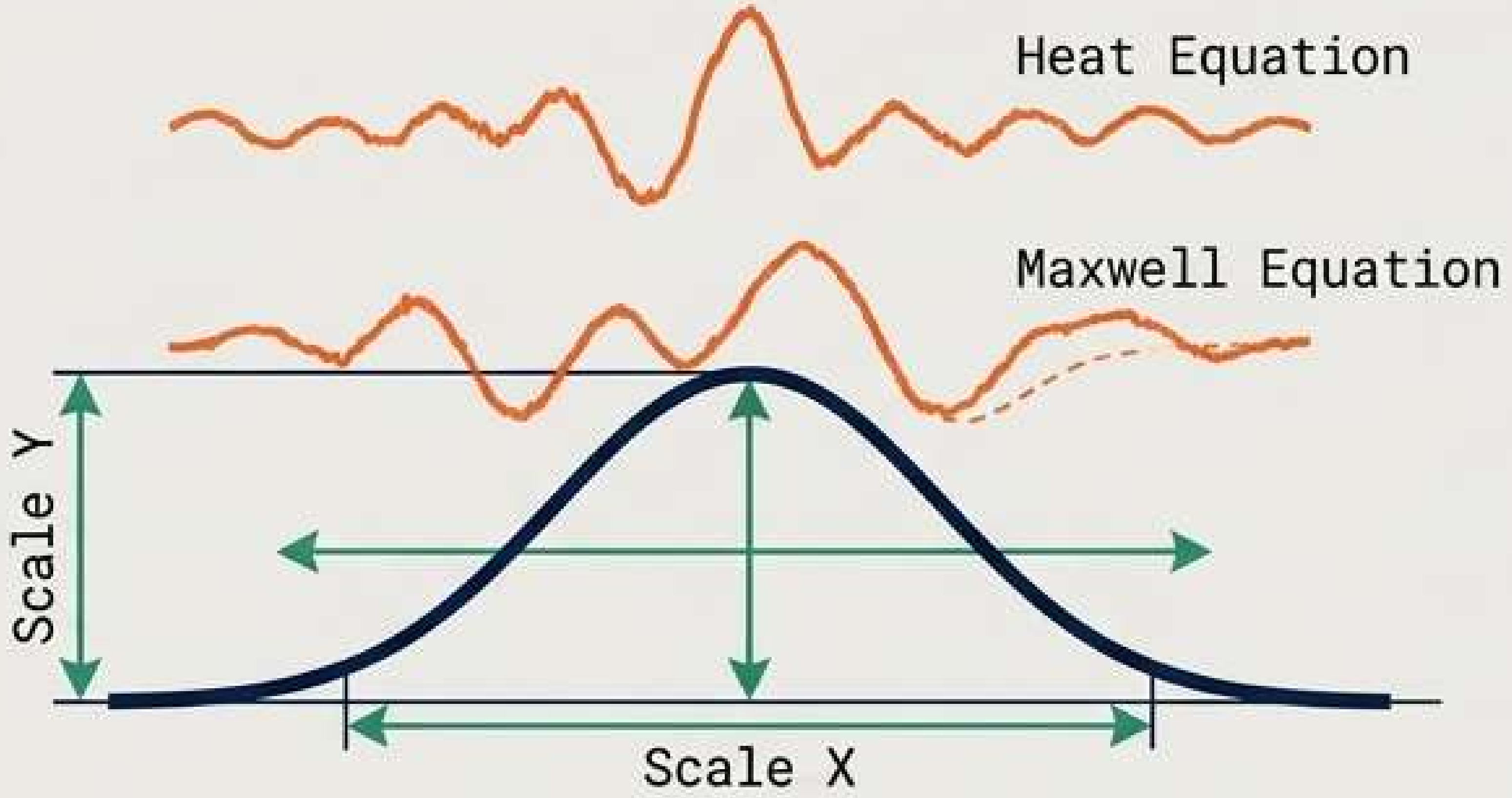
The Fundamental Constraint: Rigid Shapes in Fluid Physics

W-PINN Limitation



Hardcoded Shape.
Fast analytic derivatives, but rigid.

AW-PINN Limitation

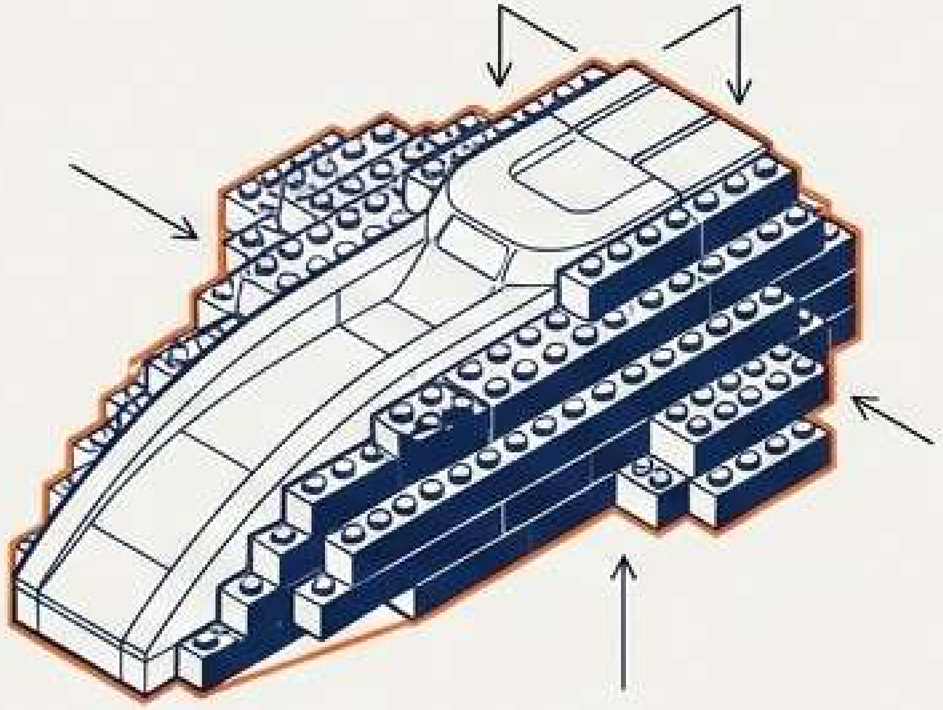


Learns Scale & Translation. Slides and resizes, but the underlying geometry is frozen. Requires manual threshold guessing.

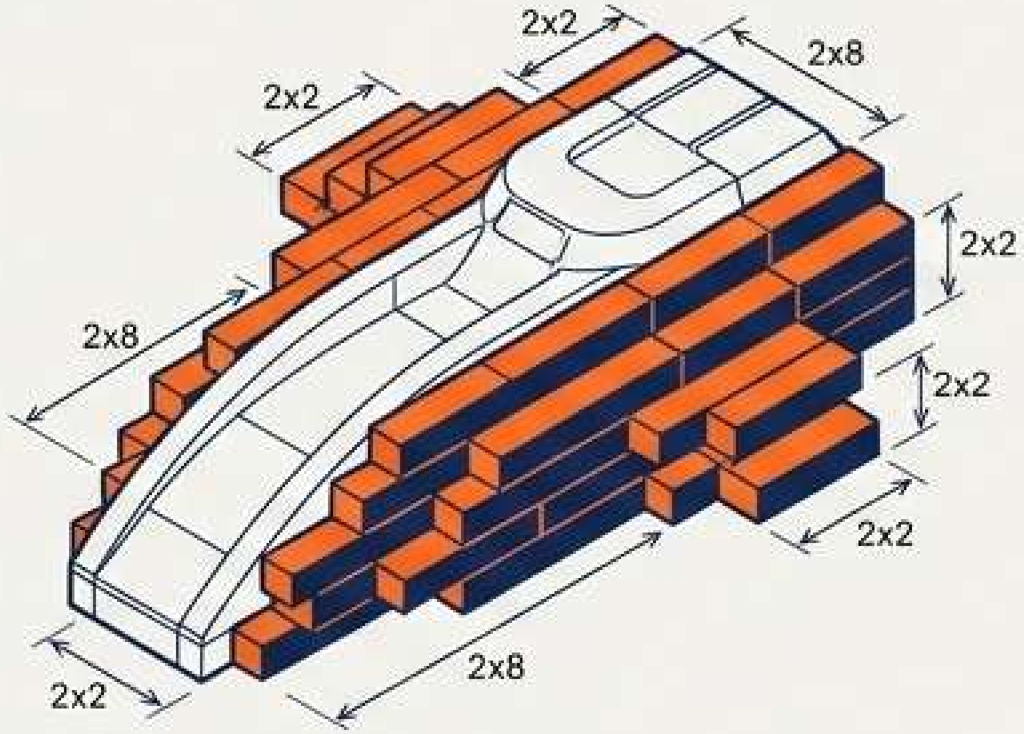
The optimal wavelet shape for a heat equation is NOT the same as the optimal shape for a Maxwell equation. Fixing the shape is an artificial constraint that hurts accuracy.

From Standard Blocks to a 3D Printer

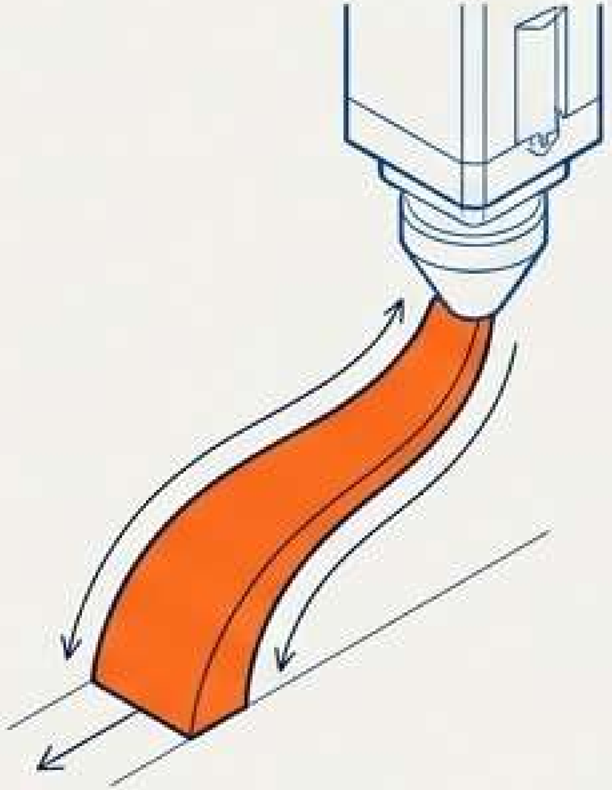
W-PINN



AW-PINN



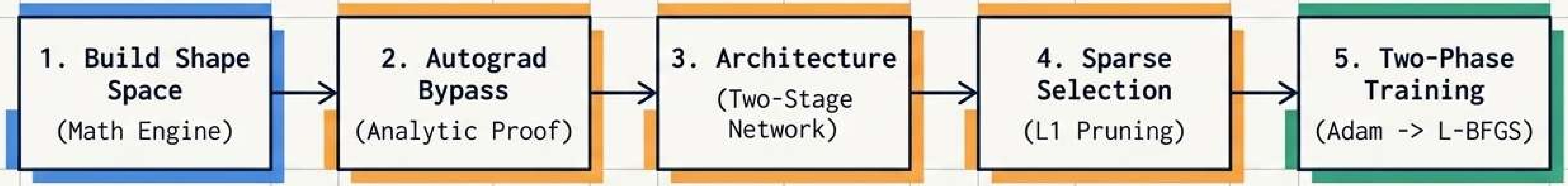
MW-PINN



Imagine trying to build a complex, curved spaceship out of Lego. W-PINN gives you stan- ndard 2x4 rectangular bricks. AW-PINN lets you resize those bricks to 2x2 or 2x8, but they remain rectangles. MW-PINN (our method) gives you a 3D printer, allowing you to invent entirely new, custom-curved pieces on the fly to perfectly fit the spaceship's hull!

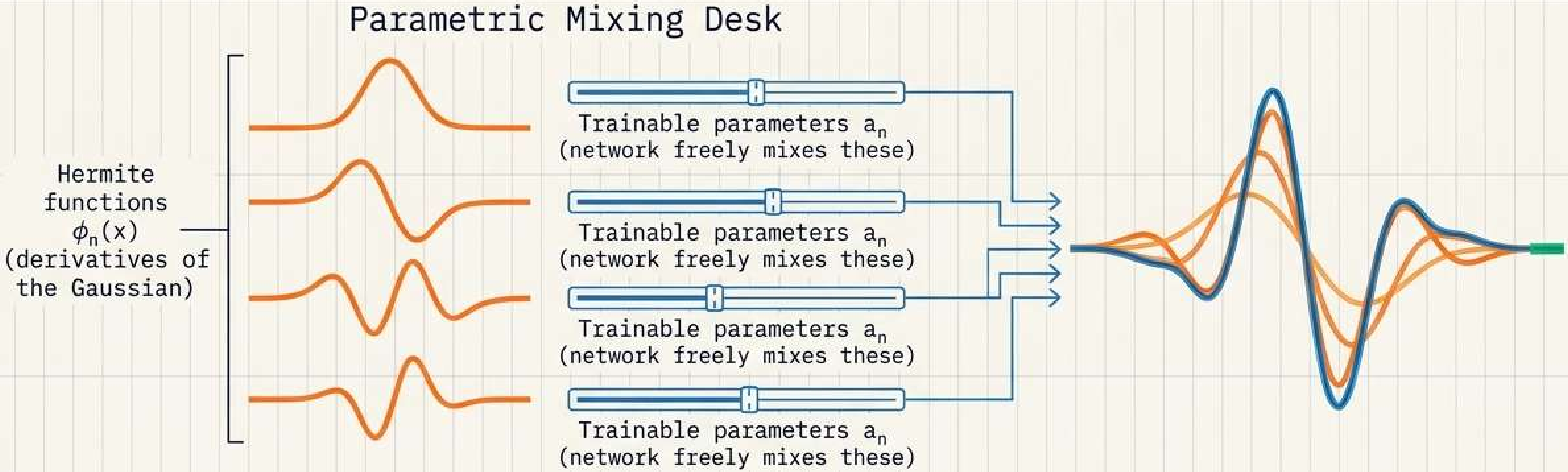
The MW-PINN Architecture: A 5-Step Blueprint

We learn the optimal mother wavelet shape itself, simultaneously with its scale and translation, using a parameterization that preserves fully analytical derivatives.



Step 1: Building a Mathematical Shape Space

$$\psi_\theta(\mathbf{x}) = \sum (a_n * \phi_n(\mathbf{x}))$$

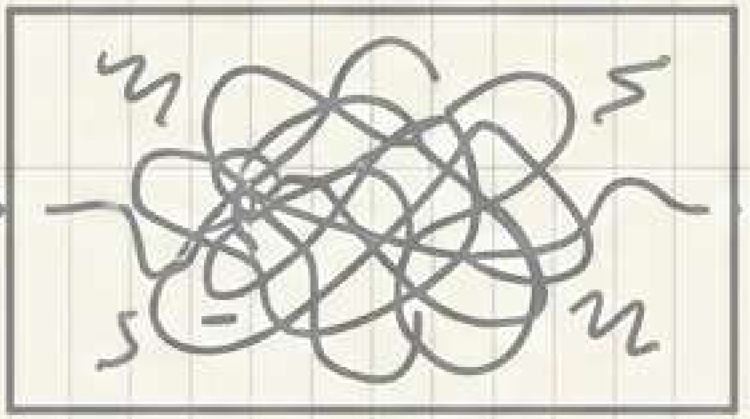


The network doesn't rely on a single, fixed bump. It discovers any weighted combination by dynamically adjusting the faders.

Step 2: The Autograd Bypass Lane

Standard PINN

Black Box Autograd



Derivative

MW-PINN

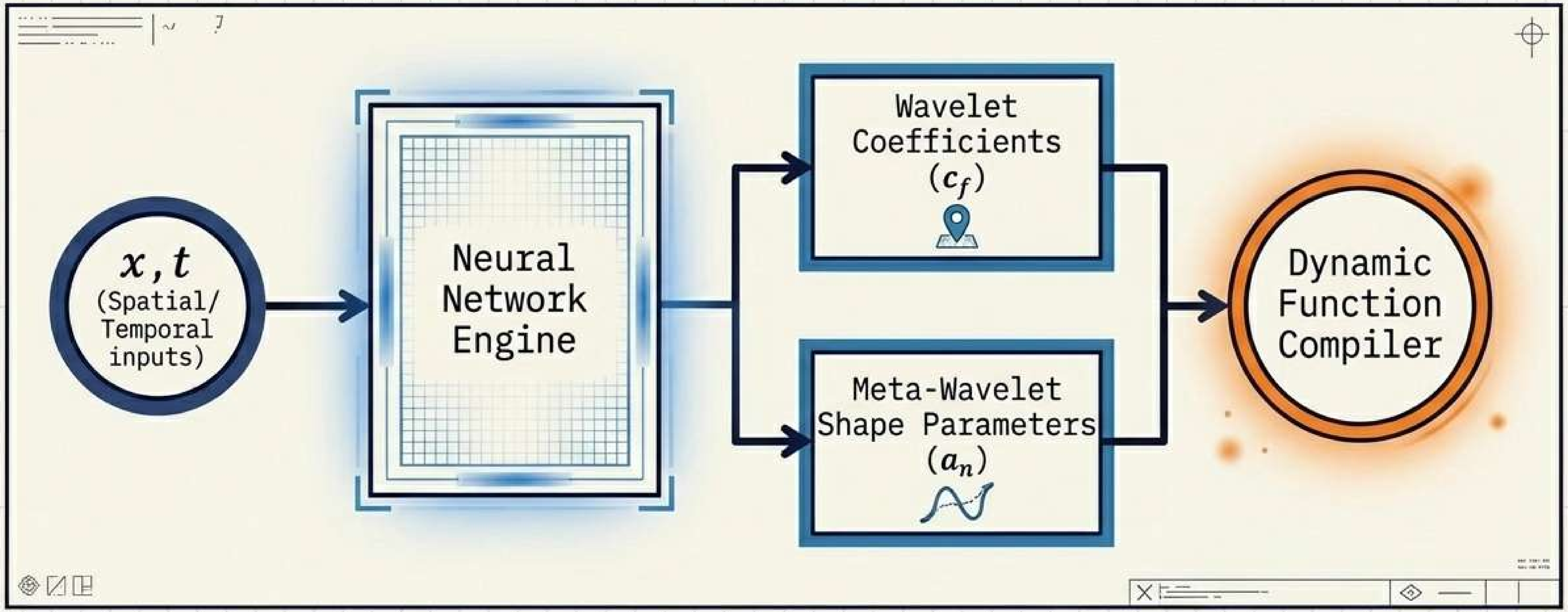
$$\frac{d^k}{dx^k}\phi_n(x) = (-1)^k\,\phi_{n+k}(x)$$

Analytical Derivative

Can we compute derivatives analytically if the shape constantly changes? Yes. Any derivative of our new meta-wavelet is simply another linear combination of Hermite functions.

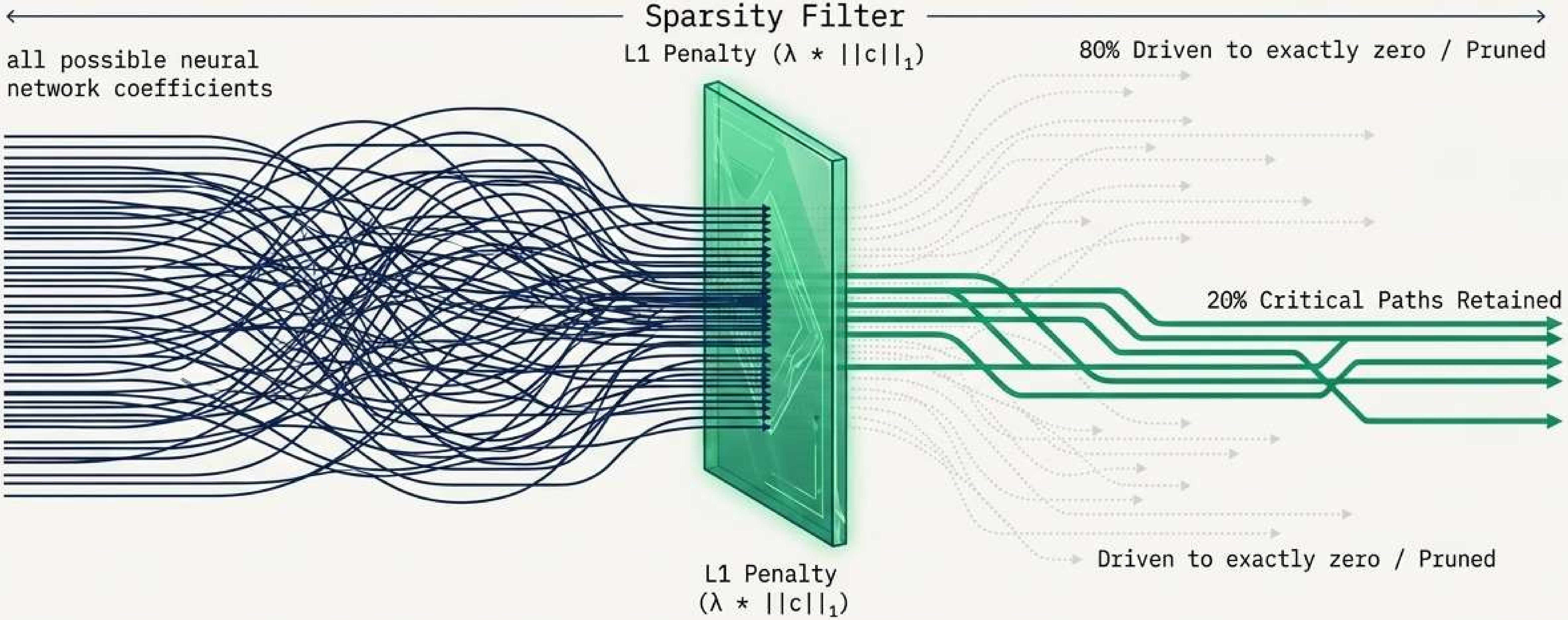
Result: Computational speed verified to essentially machine precision. No autograd needed.

Step 3: The Two-Stage Architecture



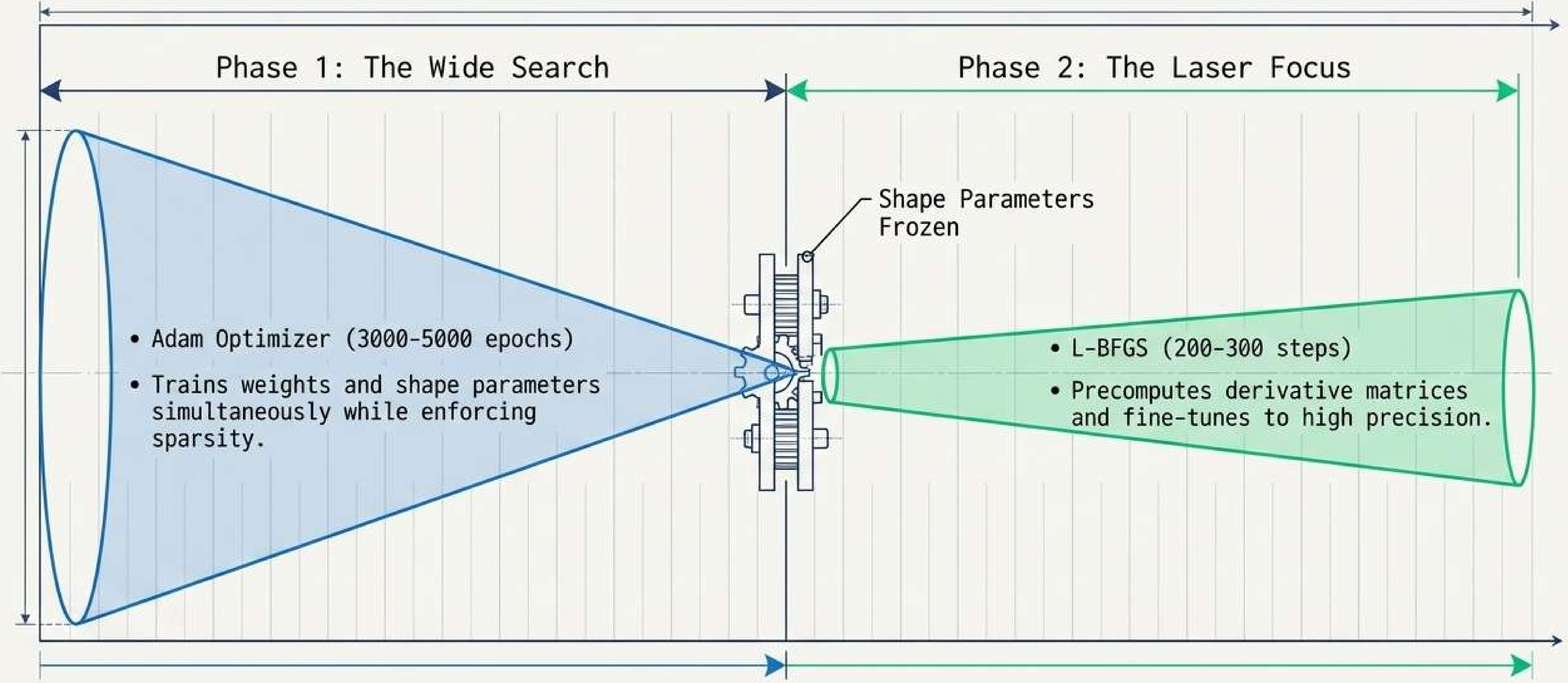
The network acts as a dual-engine: managing WHERE the wavelets go (coefficients) and WHAT they look like (shape parameters) simultaneously.

Step 4: Automatic Sparse Selection



Eliminating the manual guessing game. AW-PINN required a human-tuned score threshold. MW-PINN uses an L1 penalty to naturally force unimportant coefficients to exactly zero, automatically pruning the architecture.

Step 5: The Two-Phase Training Protocol

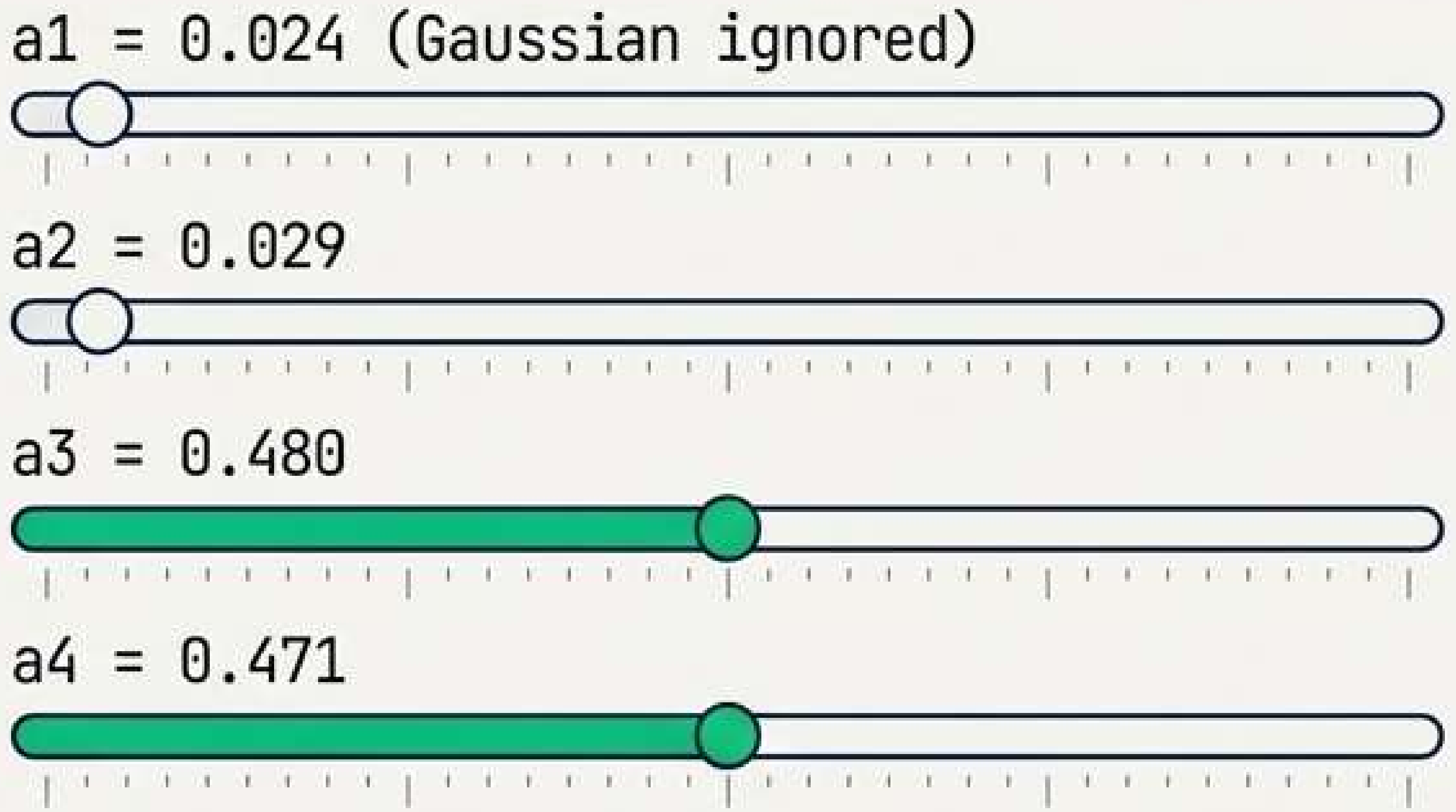


Benchmark Synthesis: The Surprising Discovery

Relative L2 Error on the Heat Conduction PDE

1	W-PINN (fixed Gaussian)	0.795
2	W-PINN + L-BFGS	0.221
3	**MW-PINN (NH=4)**	**0.169**

Visual Mixing Desk Results



When allowed to freely choose the shape with NH = 4, the network learned: a1 = 0.024, a2 = 0.029, a3 = 0.480, a4 = 0.471. It almost entirely ignored the standard shapes (a1, a2 ≈ 0) and discovered that a blend of the 3rd and 4th derivatives was optimal. This empirically proves that fixing the shape to a Gaussian is sub-optimal.

The Generational Evolution Matrix

	W-PINN	AW-PINN	MW-PINN
Learns scale/translate	No	Yes	**Yes**
Learns wavelet shape	No	No	**Yes**
Analytical derivatives	Yes	Yes	**Yes** (Bypass autograd)
Sparse selection	No	Manual	**Auto L1**
Higher accuracy	Baseline	Partial	**Yes** (1.4x improvement)
Generational Rank	1st Generation	2nd Generation	**3rd Generation** (Final)

The Paramount Workbench: Key Takeaways

1. Flexibility Wins

Letting the **neural network dynamically learn** and morph the **wavelet shape** consistently beats hardcoded standard wavelets.

2. Speed Maintained

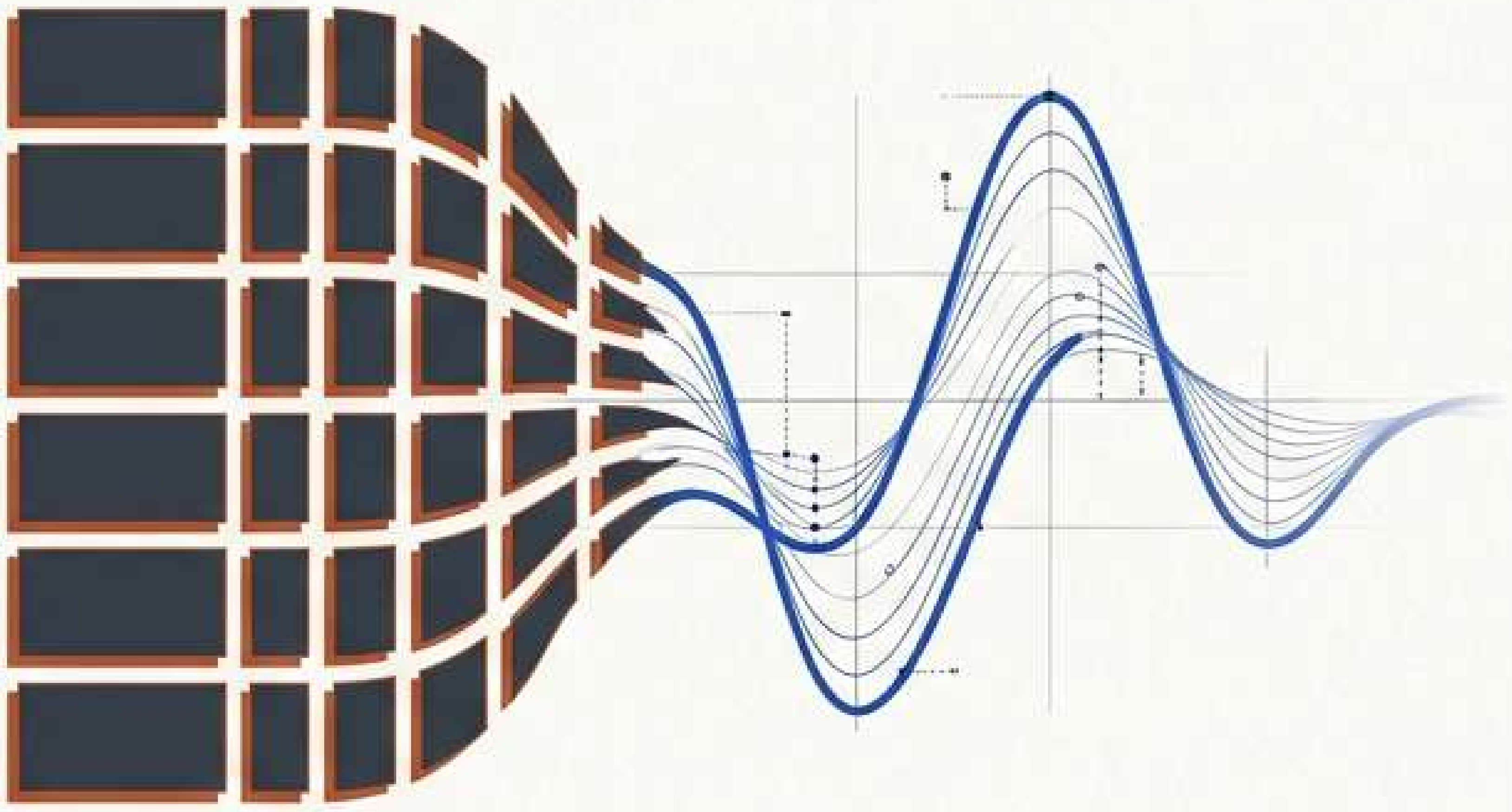
By leveraging **Hermite-Gaussian properties**, we achieved this flexibility *without* sacrificing the **fast, autograd-free training** benefits of traditional W-PINNs.

3. Automation over Tuning

Swapping **manual thresholding** for an **L1 sparsity penalty** streamlines the **pipeline**, allowing the model to **optimize its own architecture** seamlessly.

Meta-Wavelet PINN

Redefining Physics-Informed Neural Networks through Dynamic Shape Learning



OCTOBER 26, 2024

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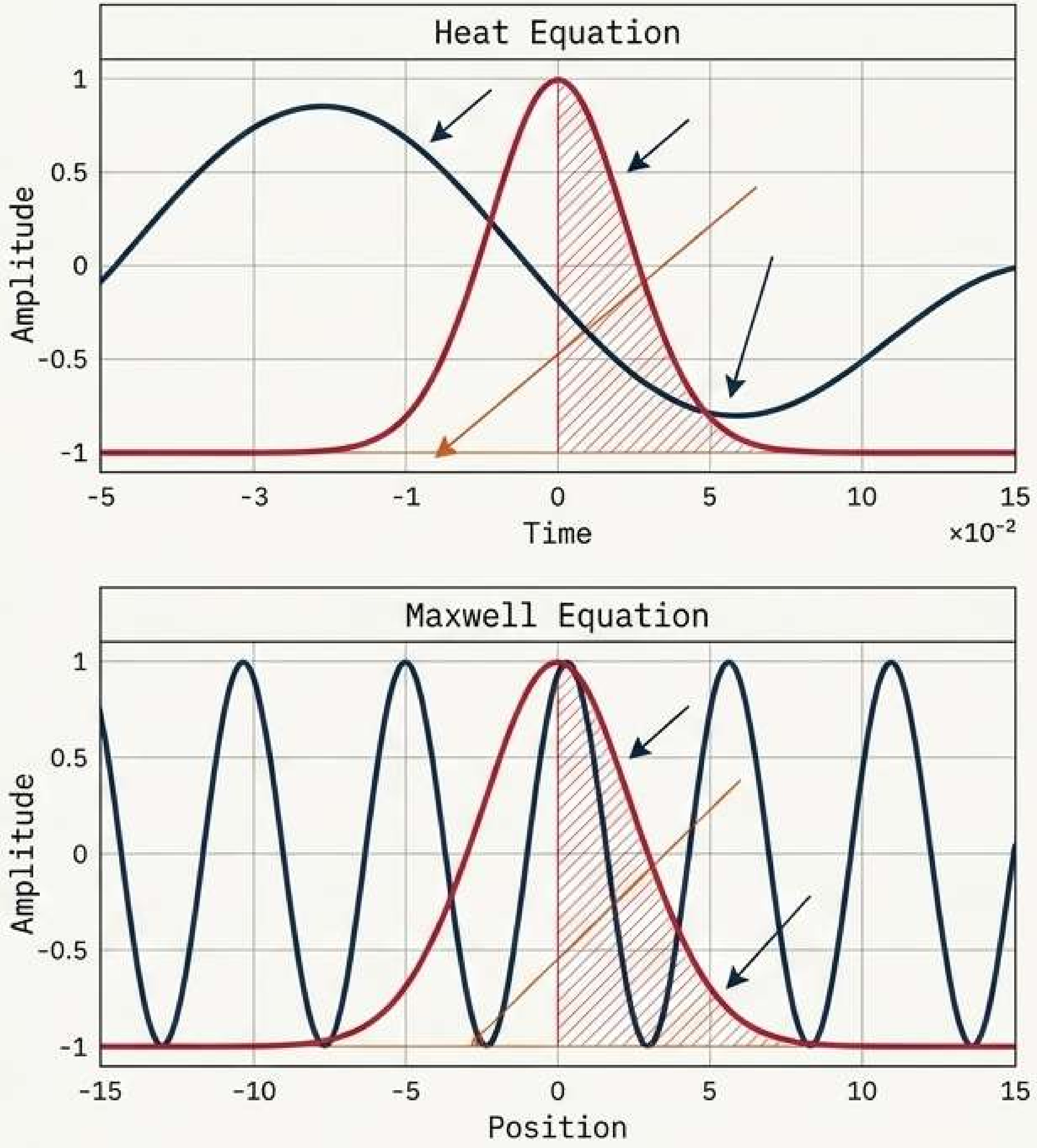
The Hardcoded Constraint

The Constraint

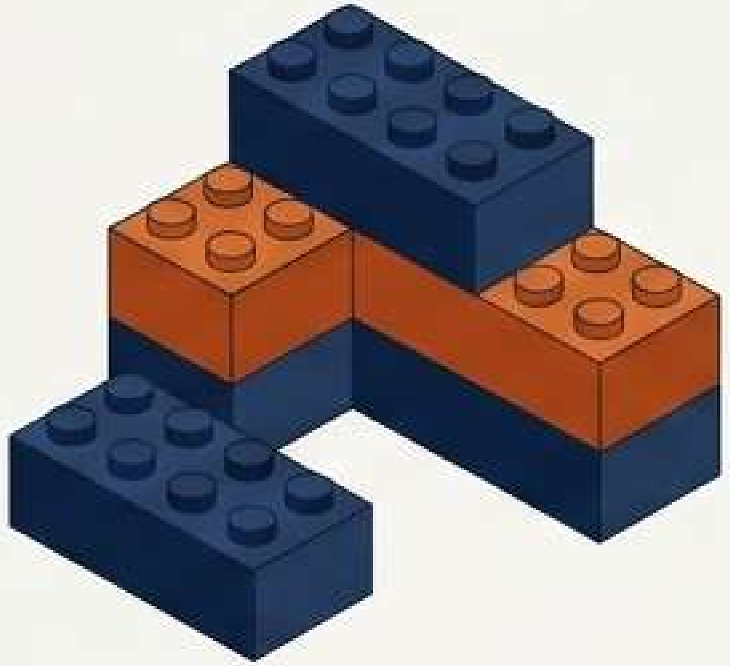
The optimal wavelet shape for a heat equation is NOT the same as the optimal shape for a Maxwell equation.

Fixing the shape is an artificial constraint that mathematically caps accuracy.

Fixing the shape is an artificial constraint that mathematically caps accuracy.

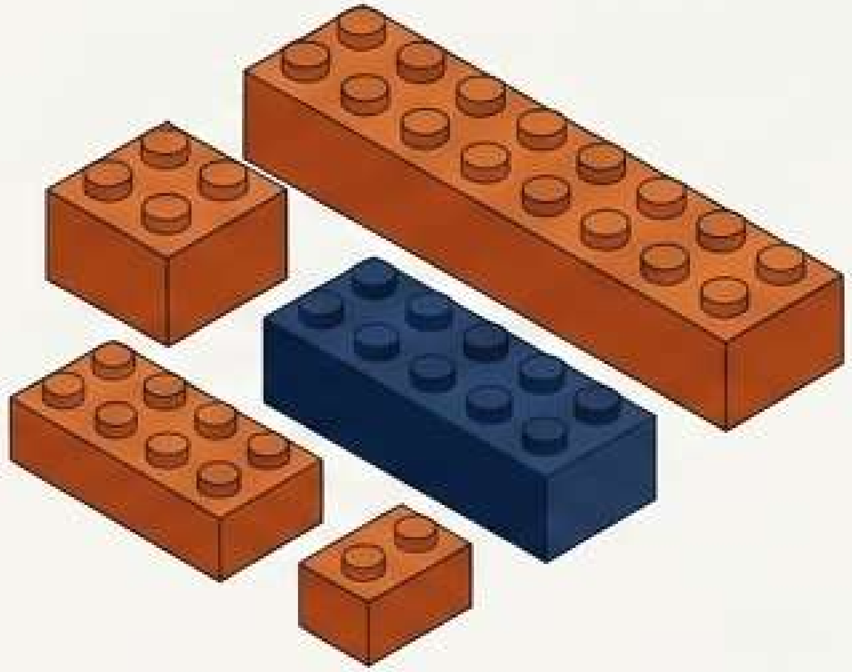


The Evolution of Wavelet PINNs



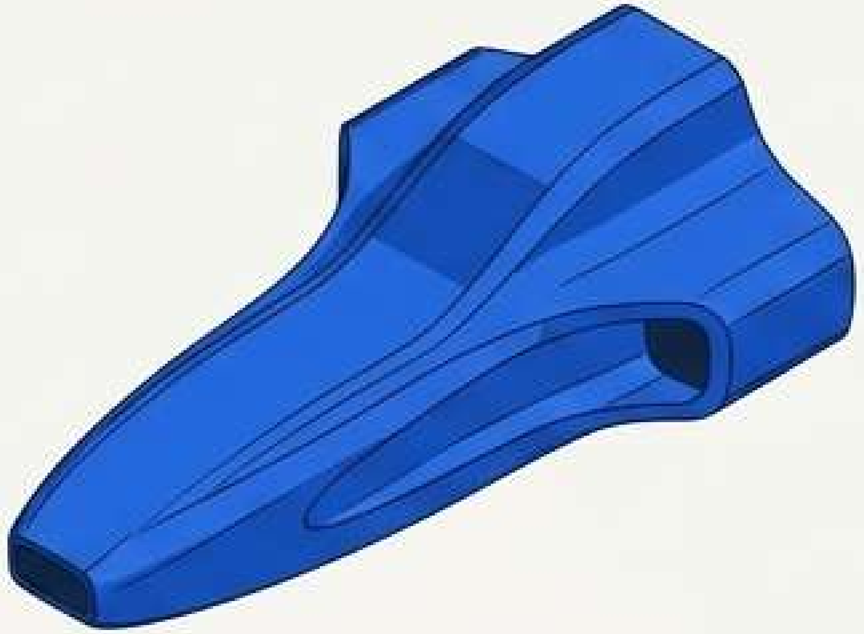
W-PINN (Fixed)

Hardcoded shape (Gaussian/Mexican Hat).
Analogy: Standard 2x4 bricks.



AW-PINN (Adaptive)

Learns scale and translation.
Analogy: Resizable bricks, but structurally restricted.



MW-PINN (Meta-Wavelet)

Learns the optimal mother wavelet shape itself.
Analogy: A 3D printer creating custom-curved pieces on the fly.

Our method, MW-PINN, functions like a 3D printer, allowing you to invent entirely new, custom-curved pieces on the fly to perfectly fit the complex, curved geometry of a spaceship's hull, unlike W-PINN and AW-PINN which offer limited, pre-defined shapes.

The MW-PINN Engine: A Learnable Shape Space

Trainable shape parameters.

Hermite functions (Gaussian derivatives).

$$\psi_\theta(x) = \sum_{n=1 \text{ to } N_H} (a_n \cdot \phi_n(x))$$

Number of basis shapes.

The Critical Proof: $\frac{d^k}{dx^k}\phi_n(x) = (-1)^k \cdot \phi_{n+k}(x)$

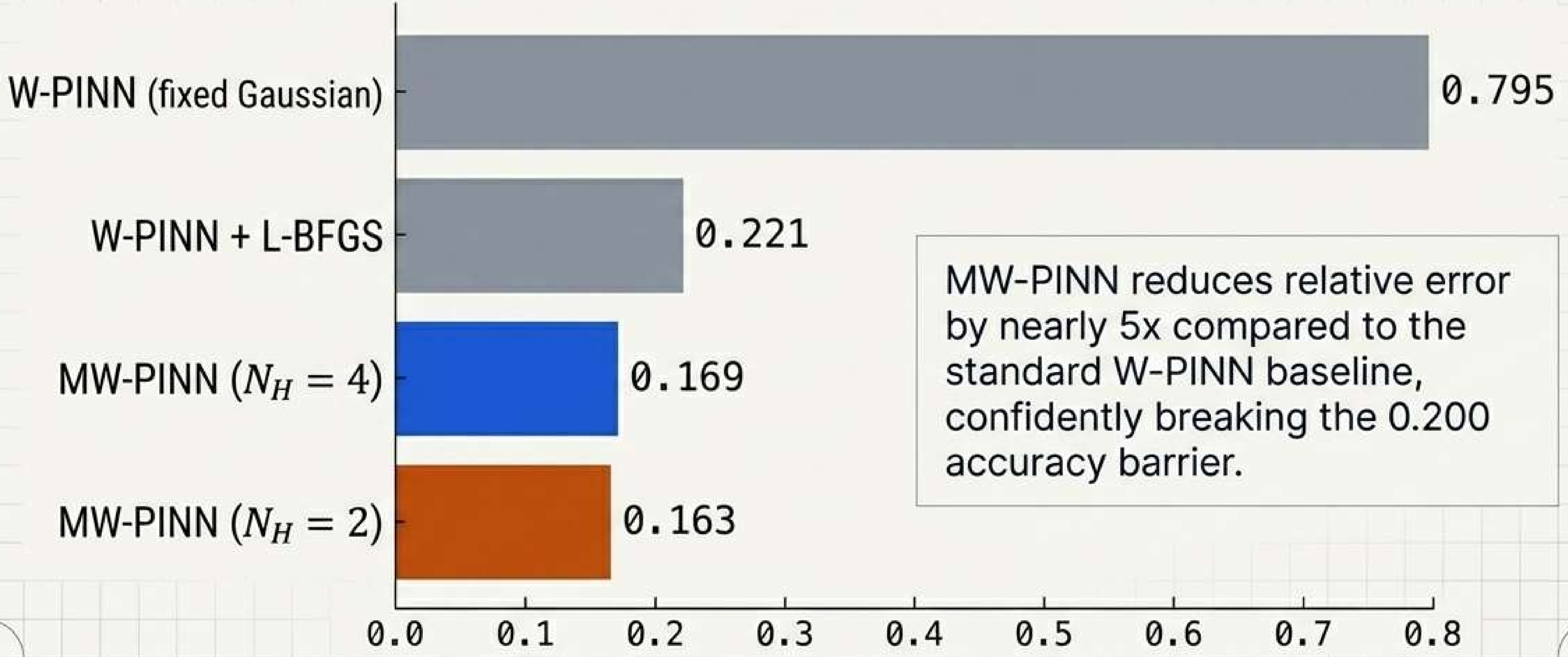
Any derivative of the meta-wavelet is just another linear combination. No autograd needed.

The Proving Ground

Benchmark: Heat Conduction PDE

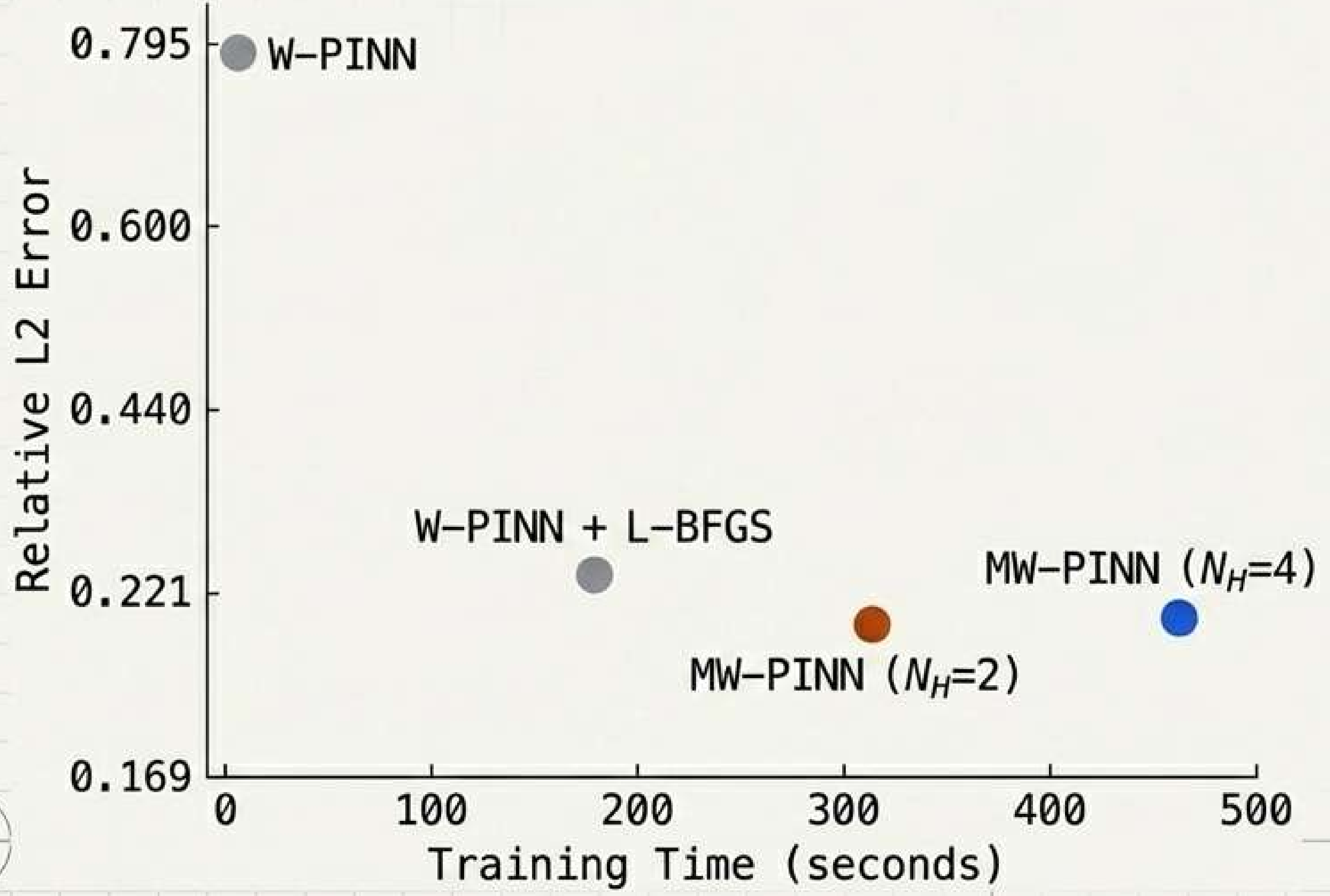
Pitting `W-PINN` baselines against `MW-PINN` across relative L2 `error` and training efficiency.

Massive Reduction in Relative L2 Error



MW-PINN reduces relative error by nearly 5x compared to the standard W-PINN baseline, confidently breaking the 0.200 accuracy barrier.

Precision Over Time



While dynamic shape learning extends training computation, it permanently escapes the mathematical accuracy floor inherent to fixed-shape models.

The Surprising Discovery

We set $N_H = 4$, giving the neural network **four** distinct Hermite-Gaussian basis shapes to freely mix.

What shape did the network learn to build?

When allowed to freely choose the shape with $N_H = 4$, the network learned:
$a_1 = 0.024$, $a_2 = 0.029$, $a_3 = 0.480$, $a_4 = 0.471$.
It almost entirely **ignored** the standard shapes ($a_1, a_2 \approx 0$) and discovered that a **blend of the 3rd and 4th derivatives** was optimal.
This empirically proves that fixing the shape to a Gaussian is sub-optimal.

Discarding the Standard

a1 = 0.024

a2 = 0.029

The network almost entirely ignored the standard base shapes ($a1,\ a2 \approx 0$). The traditional Gaussian baseline was definitively rejected by the optimizer.

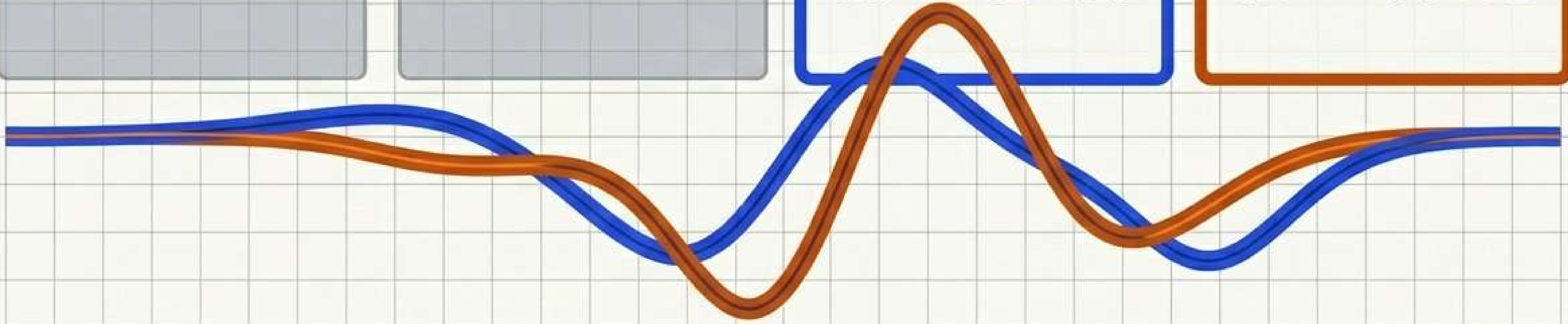
The Optimal Blend Discovered

a1 = 0.024

a2 = 0.029

a3 = 0.480

a4 = 0.471



The network discovered that a nearly equal blend of the 3rd and 4th derivatives was optimal.
This empirically proves that fixing the shape to a Gaussian is sub-optimal.

Summary vs. Prior Work

Feature	W-PINN	AW-PINN	MW-PINN
Learns scale/translate	No ✗	Yes ✓	Yes ✓
Learns wavelet shape	No ✗	No ✓	Yes ✓
Analytical derivatives	Yes ✓	Yes ✓	Yes ✓
Sparse selection	No ✗	Manual ✗	Auto L1 ✓
Higher accuracy	– ✗	Partial ✓	Yes (1.4x) ✓

Engineered for Efficiency

Two-Phase Protocol

Phase 1 (Adam)
Trains weights & shape parameters simultaneously + enforces L1 sparsity (eliminating manual thresholds).

Phase 2 (L-BFGS)
Freezes the optimal shape, precomputes derivative matrices, fine-tunes to precision.

Clean Architecture

```
|-- config.py
|-- core/
|    |-- hermite_family.py
|    |-- meta_wavelet.py
|    |-- sparse_selection.py
```

Key Conclusions

1. Flexibility Wins: Letting the neural network dynamically learn and morph the wavelet shape consistently beats hardcoded standard wavelets.

2. Speed Maintained: By leveraging Hermite-Gaussian properties, we achieved this flexibility *without* sacrificing the fast, autograd-free training benefits of traditional W-PINNs.

3. Automation over Tuning: Swapping manual thresholding for an L1 sparsity penalty streamlines the pipeline, allowing the model to optimize its own architecture seamlessly.